INCH-POUND

MIL-G-70721A (AR)
10 April 1992
SUPERSEDING
MIL-G-70721 (AR)
28 March 1988

# MILITARY SPECIFICATION

## GRENADE AND TUBE ASSEMBLY METAL PARTS ASSEMBLY

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of two component grenade and stem assembly used to generate a kill mechanism used in the M692/M731 munition. (See 6.1)

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards and handbooks. The following specifications, standards, and handbooks forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2)

SPECIFICATIONS

MILITARY

| | | |
|---|---|---|
| MIL-B-117 | - | Bags and Sleeves, Interior Packaging. |
| MIL-A-48078 | - | Ammunition, Standard Quality Assurance Provisions, General Specification For. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1320

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STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.7)

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8865569 - Shell, Lower
8865570 - Shell-Tube Assembly

(Copies of drawings, publications, and other Government documents required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation. (See 6.2)

PUBLICATIONS

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ATSM-E-18 - Standard Methods for test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

ASTM B 117 - Standard Method of Salt Spray (Fog) Testing



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(Application for copies shall be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between text of this document and the references cited herein, the text of this document shall take precedence. Nothing in the document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts and assemblies. The parts and assemblies shall comply with all requirements specified on assembly drawings and associated drawings and with all requirements specified in applicable specification.

3.3 First article testing. This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

3.4 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.



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4.1.1. Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classed as follows:

a. First article inspection (See 4.3).

b. Quality conformance inspection (See 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated in Table I.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I as specified herein.

4.3.3 Rejection. See MIL-A-48078.



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TABLE I. First article inspection

MIL-G-70721A(AR)

## CLASSIFICATION OF CHARACTERISTICS

| PARAGRAPH | TITLE<br>Grenade and Tube Assembly<br>Metal Parts Assembly | SHEET 1 OF 2 | DRAWING NUMBER<br>--- |
|---|---|---|---|
| | | | NEXT HIGHER ASSEMBLY<br>--- |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| | Sleeve, Detonator Tube (Dwg. 8864980) | | | |
| | Examination for defects | 25 | 3.2 | 4.4.2.1 |
| | Tube, Detonator (Dwg. 8865558) | | | |
| | Examination for defects | 25 | 3.2 | 4.4.2.2 |
| | Shell, Lower (Dwg. 8865569) | | | |
| | Examination for defects | 25 | 3.2 | 4.4.2.3 |
| | Stock Control | 5 | 3.2 | 4.4.3.1 |
| | Salt Spray | 25 | 3.2 | 4.4.3.2/ASTM Standard Method B 117 |
| | Minimum Inside Volume | 25 | 3.2 | 4.4.3.3/Gage |
| | Shell, Upper (Dwg. 8865568) | | | |
| | Examination for defects | 25 | 3.2 | 4.4.2.4 |
| | Stock Control | 5 | 3.2 | 4.4.3.1 |
| | Salt Spary | 25 | 3.2 | 4.4.3.2/ASTM Standard Method B 117 |
| | Minimum Inside Volume | 25 | 3.2 | 4.4.3.3/Gage |

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-G-70721A(AR)

| PARAGRAPH | TITLE<br>Grenade and Tube Assembly<br>Metal Parts Assembly | SHEET 2 OF 2 | DRAWING NUMBER<br>--- |
|---|---|---|---|
| | | | NEXT HIGHER ASSEMBLY<br>--- |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| | Shell Tube Assembly | | | |
| | (Dwg. 8865570) | | | |
| | Examination for defects | 25 | 3.2 | 4.4.2.5 |
| | Torque Test | 25 | 3.2 | Gage |
| | Push Through Test | 25 | 3.2 | Gage |
| | Tube, Detonator | | | |
| | (Dwg. 8865558) | | | |
| | Examination for defects | 25 | 3.2 | 4.4.2.6 |

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.



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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot provisions of MIL-A-48078.

a. Fastener spring grip from only one manufacturer of not more than one grade of material.

b. Steel from only one manufacturer of not more than one grade.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. Where cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.



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TABLE II Attributes sampling inspection.

| Lot Size | Inspection Level | | | |
|---|---|---|---|---|
| | I | II | III | IV |
| 2 to 8 | * | * | * | * |
| 9 to 15 | * | * | * | 13 |
| 16 to 25 | * | * | * | 13 |
| 26 to 50 | * | * | 32 | 13 |
| 51 to 90 | * | * | 32 | 13 |
| 91 to 150 | * | 125 | 32 | 13 |
| 151 to 280 | * | 125 | 32 | 32 |
| 281 to 500 | * | 125 | 32 | 32 |
| 501 to 1200 | * | 125 | 80 | 50 |
| 1201 to 3200 | 1250 | 125 | 80 | 50 |
| 3201 to 10000 | 1250 | 125 | 125 | 50 |
| 10001 to 35000 | 1250 | 315 | 125 | 80 |
| 35001 to 150000 | 1250 | 315 | 125 | 80 |

Numbers under inspection levels indicate sample size; asterisks indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.5). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractors quality program or detailed inspection system, as applicable.



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# QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF CHARACTERISTICS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.2.1 | Sleeve, Detonator Tube | 1 OF 1 | 8864980 |
| | | | NEXT HIGHER ASSEMBLY 8865570 |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined | | | |
| Major | | | | |
| 101 | Diameter of body | Level II | 3.2 | Gage |
| 102 | Burr(s) | Level II | 3.2 | Visual |
| 103 | Crack or perforation | Level II | 3.2 | Visual |
| 104 | Inside diameter | Level II | 3.2 | Gage |
| 105 | Finish | | 3.2 | Certification |
| 106 | Shoulder length | Level II | 3.2 | Gage |
| 107 | Shoulder diameter | Level II | 3.2 | Gage |
| Minor | | | | |
| 201 | Poor workmanship | Level IV | 3.5 | Visual |

NOTES:



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## CLASSIFICATION OF CHARACTERISTICS



| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.2 | Tube, Detonator | 1 OF 1 | 8865558 |
| | | | NEXT HIGHER ASSEMBLY: 8865570 |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined | | | |
| Major | | | | |
| 101 | True position of flash hole | Level II | 3.2 | Gage |
| 102 | Diameter of flash hole | Level II | 3.2 | Gage |
| 103 | Pitch diameter of thread | Level II | 3.2 | Gage |
| 104 | Diameter of cavity above thread, minimum | Level II | 3.2 | Gage |
| 105 | True position of diameter of cavity above thread with outside diameter | Level II | 3.2 | Gage |
| 106 | True position of smaller (sleeve) cavity with outside diameter | Level II | 3.2 | Gage |
| 107 | Length of effective thread, min. | Level II | 3.2 | Gage |
| 108 | Angle missing from flash hole | Level II | 3.2 | Visual |
| 109 | Flash hole missing | Level II | 3.2 | Visual |
| Minor | | | | |
| 201 | Minor diameter of thread, max. | Level IV | 3.2 | Gage |
| 202 | Diameter of thread counterbore | Level IV | 3.2 | Gage |
| 203 | Depth of thread counterbore | Level IV | 3.2 | Gage |
| 204 | Poor workmanship | Level IV | 3.5 | Visual |

NOTES:



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# QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF CHARACTERISTICS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.3 | Shell, Lower | 1 OF 1 | 8865569 |
| | | | NEXT HIGHER ASSEMBLY: --- |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined | | | |
| Major | | | | |
| 101 | Largest outside diameter | Level II | 3.2 | Gage |
| 102 | Thickness, min. | Level II | 3.2 | Gage |
| 103 | Profile improper | Level II | 3.2 | Gage |
| 104 | Diameter of hole | Level II | 3.2 | Gage |
| 105 | Concentricity of hole with largest outside diameter to mating surface perpendicularity | Level II | 3.2 | Gage |
| 106 | Angle incorrect on lip end | Level II | 3.2 | Visual |
| 107 | Stock control | | 3.2 | 4.4.3.1 |
| 108 | Salt spray | | 3.2 | 4.4.3.2/ASTM Standard Method B 117 |
| 109 | Minimum inside volume | 10/0/1 | 3.2 | 4.4.3.3/Gage |
| 110 | Diameter of largest inside diameter at basic height | See Note 1 | 3.2 | 4.4.3.4/Gage |
| 111 | True position of largest inside diameter at basic height | See Note 1 | 3.2 | 4.4.3.4/Gage |
| Minor | | | | |
| 201 | Small lip radius missing | Level IV | 3.2 | Visual |
| 202 | Poor workmanship | Level IV | 3.5 | Visual |

NOTES:
Note 1 - Except second sentence should read, "In addition, a random sample of five(5) parts shall be selected from the last portion of each four(4) hour's production..." applies.



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## CLASSIFICATION OF CHARACTERISTICS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.4 | Shell, Upper | 1 OF 1 | 8865568 |
| | | | NEXT HIGHER ASSEMBLY: --- |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined | | | |
| Major | | | | |
| 101 | Distance from hole to flat end including chamfer | Level II | 3.2 | Gage |
| 102 | Largest Outside diameter | Level II | 3.2 | Gage |
| 103 | Diameter of hole, min. | Level II | 3.2 | Gage |
| 104 | Depth to hole face, min. | Level II | 3.2 | Gage |
| 105 | True position of hole with outside diameter | Level II | 3.2 | Gage |
| 106 | True position of second largest outside diameter at basic height | Level II | 3.2 | Gage |
| 107 | Thickness, min. | Level II | 3.2 | Gage |
| 108 | Profile improper | Level II | 3.2 | Gage |
| 109 | Angle of outside diameter incorrect | Level II | 3.2 | Gage |
| 110 | Stock control | | 3.2 | 4.4.3.1 |
| 111 | Salt spray | | 3.2 | 4.4.3.2/ASTM Standard Method B 117 |
| 112 | Minimum inside volume | 10/0/1 | 3.2 | 4.4.3.3/Gage |
| 113 | Diameter of second largest outside diameter at basic height | See Note 1 | 3.2 | 4.4.3.4/Gage |
| Minor | | | | |
| 201 | Poor workmanship | Level IV | 3.5 | Visual |

NOTES: Note 1 - except second sentence should read, "In addition, a random sample of five(5) parts shall be selected from the last portion of each four(4) hour's production ..." applies.

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.



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## CLASSIFICATION OF CHARACTERISTICS

MIL-G-70721A(AR)

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.5 | Shell Tube Assembly | 1 OF 1 | 8865570 |
| | | | NEXT HIGHER ASSEMBLY: --- |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined | | | |
| Major | | | | |
| 101 | Torque test | Level II | 3.2 | Gage (See Note) |
| 102 | Protrusion of tube assembly | Level II | 3.2 | Gage |
| 103 | True position of diameter of tube assembly with outside diameter of upper shell including perpendicularity with shell face | Level II | 3.2 | Gage |
| 104 | Inside diameter of spring grip fastener, max. | Level II | 3.2 | Gage |
| 105 | Inside diameter of detonator tube sleeve, min. | Level II | 3.2 | Gage |
| 106 | Distance between shell hemisphere face and tube face | Level II | 3.2 | Gage |
| 107 | Component or operation missing | Level II | 3.2 | Visual |
| 109 | Push through test | 125/1/2 | 3.2 | Gage |
| Minor | | | | |
| 201 | Poor workmanship | Level IV | 3.5 | Visual |

NOTES:
1. Alternate method shown in 4.4.3.5.

AMSMC Form 1570b, 1 Jul 89

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4.4.3 Testing.

4.4.3.1 Dimensional inspection of stock specimen. Two (2) stock specimens shall be taken from each coil of finished stock just prior to forming the upper and lower shells. The width of each specimen shall be the width of the stock, and the length shall be as necessary to obtain the total surface area of the tool to be used in production. One (1) specimen shall be taken from the beginning of the coil and the other specimen shall be taken from the end of the coil. An additional specimen shall be taken whenever a new, repaired, or altered tool is entered into production. Each specimen shall be identified with an index position of the tool and for the direction of travel of the stock with respect to the tool. The specimens shall also be identified as being from the beginning or end of the coil. If, for any specimen, the average value of fifteen (15) measurements for the total thickness dimension and the depth dimension fails to comply, or if the direction of tool travel does not comply with the applicable drawing requirements, the coil, or portion thereof, from which the specimen was taken, shall be rejected. In addition, the tool which produced the rejected specimen shall be removed from production. All shells produced from the rejected coil shall be rejected. The dimensional inspection shall be performed using equipment in accordance with 4.4.4.

The specimen from a new, repaired, or adjusted tool entering production shall be subjected to dimensional inspection of all stock dimensions specified in the applicable drawings. Fifteen (15) readings shall be taken for each dimension. The tool shall not be used in production until dimensional compliance with the drawing requirements has been determined from the specimen produced by the tool.

4.4.3.2 Salt spray test of upper shell and lower shell.

4.4.3.2.1 Phosphate - coated (batch-type process). Fifteen (15) each of upper shell and lower shell shall be selected from each batch and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the batch represented by the failing part shall be rejected.

4.4.3.2.2 Phosphate - coated shells (continuous-type process). Fifteen (15) each of upper shell and lower shell shall be selected from each four (4) hours of production, or portion thereof, and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specification requirement, the quantity represented by the failing part shall be rejected.



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4.4.3.2.3 Phosphate - coated shells (lot plan). Twenty-five (25) each of upper shell and lower shell shall be selected from each lot and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the lot shall be rejected.

4.4.3.2.4 Plated shells (when applicable). Twenty-five (25) each of upper shell and lower shell shall be selected from each lot and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the lot shall be rejected.

4.4.3.3 Minimum inside volume of hemispheres (prior to phosphating). A sample of ten (10) each upper shells and lower shells shall be selected for determination of minimum volume requirement to qualify the production process. In addition, five (5) parts shall be selected from the last portion of each four (4) hours production or fraction thereof, for determination of minimum volume requirement as a control of the production process. This determination shall be performed using equipment in accordance with 4.4.4. If any defective parts are found during qualification of the production process, the process producing the defective part shall be removed from production. Further, that portion of production since the last production process control check shall be returned to the contractor for corrective action.

All production processes removed from production because of some fault may, after corrective action, be returned to production, providing they pass the qualification test above. The contractor may request a change of inspection frequency, providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.4.3.4 Dimensional control of stamped parts. In place of the normal sampling associated with the classification of defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.



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All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.4.3.5 Minimum torque of tube assembly. In place of the normal sampling procedure associated with the classification of defects, and with the approval of the Contracting Officer, the following inspection procedures may be used: at the start of each day, and whenever a new piece of equipment is put into operation, five (5) tube assemblies shall be selected from each piece of equipment for determination of minimum assembly torque of the tube assembly for qualifying the equipment. In addition, five (5) assemblies per machine shall be selected from each hour's production, or fraction thereof, for the determination of the minimum tube assembly torque as a control of the equipment. If any defective assemblies are found during qualification of the equipment, the equipment producing the defective assemblies shall not be used in production. If any defective assembly is found when inspection is performed for control of the equipment, the equipment having produced the defective assembly shall be removed from production.

Further, that portion of production since the last equipment check shall be returned to the contractor for corrective action. All equipment removed from production because of some fault may, after corrective action, be returned to production providing it passes the qualification test above.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the tables beginning with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.4 herein and MIL-A-48078, Section 6.

4.5 Methods of inspection.

4.5.1 Dimensional inspection of stock specimens. Fifteen (15) measurements shall be taken on each sample specimen for the total thickness dimension and the depth dimension. There shall be three (3) points of measurement across the width of the specimen at five (5) places along the length. The spacing of the readings shall be approximately uniform. One measurement for each characteristic shall be performed at each of the fifteen (15) measuring points on the specimen. An average value for each characteristic shall be computed for each specimen for which these measurements were obtained. In addition to the above, stock specimens from a new, repaired, or adjusted tool entering production shall have the following characteristics measured; groove radius, spherical radius, and the .0965 dimension.



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5. PACKAGING

5.1 Preservation (Level C).

5.1.1 Lower shell (Dwg. D-8865569). The lower shell shall be packed in accordance with MIL-STD-1169, except that a Type 1, Class E, Spec MIL-B-117 barrier bag and an outer fiber box shall be used in addition to the inner fiber box. The inner fiber box corners shall be blunted before placing in bag. Prior to sealing bag, exhaust all excess air. Maximum moisture content of inner fiber box and any other packaging material used within the inner box shall be 9%.

5.1.2 Shell-tube assembly (Dwg. D-8865570). Packaging shall be in accordance with 5.1.1.

5.2 Packing (Level C).

5.2.1 Lower shell (Dwg. D-8865569). Packing shall be in accordance with 5.1.1.

5.2.2 Shell-tube assembly (Dwg. D-8865570). Packing shall be in accordance with 5.1.1.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Shipping. When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The shell and shell assemblies covered by this specification is intended for use in the munition for projectiles, H.E., 155MM, M692 and M731.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced in section 2.1.



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6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078). Equipment designs shall be submitted as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAT-I (D), Picatinny Arsenal, New Jersey 07801-5000. Request letter of submittal should state contractor, contract number, specification number, item nomenclature, and classification of defects and tests paragraph number. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of test data and data card. In addition to the normal distribution of records, when the projectiles are procured by the U.S. Amry Armament Material Readiness Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAT-M (D), Picatinny Arsenal, New Jersey 07801-5000.

6.5 Submission of alternative quality conformance procedures. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor to the procuring contracting officer.

6.6 Subject term (key word) listing.
Tube, Detonator
Shell and Tube Assembly
Salt Spray
Hardness Test

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as, US Army Armament Research and Development Command, Edgewood Arsenal, Frankfort Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1320-AC80)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-G-70721A (AR) | 10 April 1992 |

3. DOCUMENT TITLE
GRENADE AND TUBE ASSEMBLY METAL PARTS ASSEMBLY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code) | |
|---|---|---|
| US ARMY ARDEC<br>STANDARDIZATION OFFICE | (1) Commercial<br>(201) 724-6675 | (2) AUTOVON<br>880-6675 |
| c. ADDRESS (Include Zip Code)<br>ATTN: SMCAR-BAC-S<br>PICATINNY ARSENAL,NJ 07806-5000 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

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